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Studierichting: Informatica
Oefeningenreeks 3H nr. 8.1

Schets de functie: $r = 1 + \cos(\theta)$

$$\left.\begin{array}{l} \operatorname{dom} r = \mathbb{R} \\ P = \pi \end{array}\right\} \text{Bep. Dom.} = [0, \pi]$$

Nulpunten:

$$r = 0 \Leftrightarrow \sin(2\theta) = 0 \Leftrightarrow 2\theta = k\pi \text{ met } k \in \mathbb{N} \longrightarrow \theta \in \left\{0, \frac{\pi}{2}, \pi\right\}$$

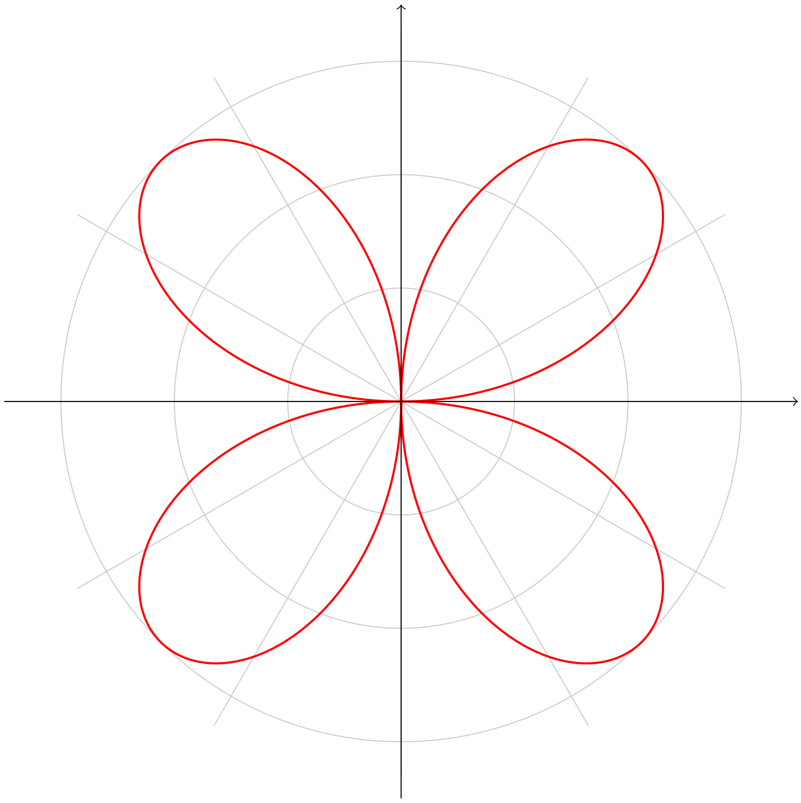
$$r' = 0 \Leftrightarrow 6\sin(2\theta) = 0 \Leftrightarrow \cos(2\theta) = 0 \Leftrightarrow 2\theta = \frac{\pi}{2} + k\pi \text{ met } k \in \mathbb{N}$$

$$\longrightarrow \theta \in \left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$$

Schema:

θ	0		$\frac{\pi}{4}$		$\frac{\pi}{2}$		$\frac{3\pi}{4}$		π
r	0	$+$	3	$+$	0	$-$	-3	$-$	0
r'	6	$+$	0	$-$	-6	$-$	0	$+$	6
	$\nearrow$	$\nearrow$	Max	$\searrow$	$\searrow$	$\searrow$	min	$\nearrow$	$\nearrow$

Grafiek in poolcoördinaten:



References